

Technology Stack

Date	5 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	React.js, TypeScript, Tailwind CSS	React provides a component-based architecture for building a responsive and interactive UI. TypeScript improves code quality and maintainability. Tailwind CSS enables fast, consistent, and responsive styling.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, FastAPI	FastAPI is a modern, high-performance framework for building APIs with automatic validation and documentation. Python offers rich libraries for ML, NLP, and audio processing.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	PostgreSQL	PostgreSQL is a reliable, open-source relational database that ensures data integrity, strong relational mapping, and supports complex queries for storing users, evaluations, reports, and metadata.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	AWS (EC2, S3, RDS, ECS), Docker	AWS provides scalable compute (EC2), secure storage (S3), and managed database (RDS). Docker ensures consistent packaging and easy deployment across environments.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub, GitHub Actions	GitHub is used for version control and collaboration. GitHub Actions automates testing, linting, and deployment pipelines for reliable and efficient CI/CD.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Firestore (Auth & Notifications), OpenAI Whisper API, Google Cloud Speech-to-Text API	Firestore handles user authentication and real-time notifications. Whisper API is used for high-accuracy audio transcription. Google Speech-to-Text API provides fallback/alternative transcription with high reliability.